

Interim Report

Data Analytics Capstone

**Data-Driven Analysis and Predictive Modeling of Graduate Employment and
Salary Outcomes in Singapore**

Interim Report

Kumaran Sundaram

Walsh College

QM640: Data Analytics Capstone

Mentor: Prof. Sharath Srivatsa

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GitHub Repository and Project Artifacts

GitHub repository for evaluator access:

https://github.com/sundaramkumaran/QM640_GES_Capstone

The repository is intended to provide access to the raw dataset, processed datasets, analysis notebook, output tables, and report figures. The local analysis package generated from the notebook is stored under the project outputs folder.

Key local paths used in the notebook are listed below. These paths should be updated if the project is moved to another computer or repository folder.

Table 1 Project artifact locations

Artifact	Path
Raw data	data/raw/graduate_employment_survey.csv
Processed data	data/processed/ges_cleaned.csv
Clustered processed data	data/processed/ges_clustered_program_profiles.csv
Notebook	notebooks/QM640_GES_Interim_Master_Analysis.ipynb
Output tables	outputs/tables/
Output figures	outputs/figures/
Report package	outputs/report_package/
Full checkpoint	checkpoints/checkpoint_full_analysis_completed.pkl

Introduction

Background and Context

Graduate employment outcomes are an important public indicator of how effectively higher education programs connect students with labour-market opportunities. In Singapore, the Graduate Employment Survey (GES) provides program-level information on employment rates and salary outcomes across autonomous universities. The dataset is useful for students comparing degree pathways, universities reviewing program outcomes, and policymakers monitoring labour-market alignment.

The project focuses on official program-level GES data for Nanyang Technological University (NTU), National University of Singapore (NUS), Singapore Institute of Technology (SIT), Singapore Management University (SMU), Singapore University of Social Sciences (SUSS), and Singapore University of Technology and Design (SUTD). The dataset is distributed through data.gov.sg by the Ministry of Education (MOE). The current analysis uses the available program-year records from 2013 through 2024.

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Graduate outcomes should not be interpreted using salary alone. A program can have a high median salary but wider salary dispersion, lower stable employment, or greater variation across years. Similarly, a program can show high overall employment but lower full-time permanent employment. These differences matter because students and institutions often need to distinguish general employment from stable employment and salary level from salary consistency.

This interim report presents a complete analytical workflow from data preparation through exploratory analysis, statistical testing, predictive modelling, leakage-controlled validation, and clustering. The report is written as a standalone document so that a reader can understand the project status without referring to the earlier synopsis.

Problem Statement

Graduate employment datasets are commonly viewed through isolated indicators such as median salary or overall employment rate. This creates an interpretation gap because graduate outcomes are multidimensional. Employment stability, salary spread, salary growth over time, university context, and program-level differences may all affect the value of a degree program. This project addresses the gap by creating an integrated analytics framework that explains, predicts, and segments program-level employment and salary outcomes using official Singapore GES data.

Purpose of the Study

The purpose of this study is to analyze Singapore university program-level graduate outcomes using descriptive, inferential, predictive, and segmentation methods. The study aims to identify employment and salary trends, test whether outcomes differ across universities, examine factors associated with salary and Career Value Index, classify high-outcome program-year records, and segment programs into meaningful employability-salary profiles.

Response to Synopsis and Mentor Feedback

The synopsis evaluation feedback indicated that the project scope and objectives should be explained more clearly in layman terms, and that the problem statement, objectives, and solutioning should be extended into more measurable outcomes. The feedback also noted that the research questions should be more clearly linked to variables from the data dictionary and to the proposed analytical solution.

In response, the interim report reframes the project as a measurable analytics solution rather than only a descriptive or theoretical study. Each research question is now linked to specific data variables, analytical methods, measurable outputs, and capstone deliverables. The report also

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includes sample-size adequacy checks by research question, an expanded exploratory data analysis, statistical testing, regression modelling, leakage-controlled classification, clustering-based segmentation, and a Career Value Index as a decision-support contribution.

Following mentor feedback, the interim report has been strengthened in three areas. First, the research-question findings are summarized using a response and technical evidence table.

Second, the Career Value Index is validated using correlation analysis with p-values, diagnostic regression interpretation, and planned sensitivity checks. Third, detailed EDA outputs are provided in Appendix B so that the main report focuses more directly on findings, interpretation, and decision-support value.

Interim Project Status (Progress Snapshot)

- Completed: data loading, cleaning, missing-value audit, feature engineering, dataset overview, data dictionary, and full exploratory data analysis.
- Completed: statistical testing for university-level differences, salary regression, Career Value Index diagnostic modelling, preliminary classification, and leakage-controlled classification refinement.
- Completed: K-Means clustering and program segmentation for RQ5.
- Completed: final research-question summary, checkpoint save, report package export, and report-ready narrative text generation.
- Completed: Career Value Index validation using Pearson correlation, Spearman correlation, and p-values to show statistically significant relationships with relevant graduate outcome features.
- Although this is an interim submission, the analysis has progressed beyond basic exploratory data analysis and includes preliminary inferential, predictive, leakage-controlled, and segmentation results that will be refined further in the final report.
- Pending for final report: complete CVI sensitivity analysis, refine coefficient interpretation, report selected pairwise university differences, polish visual presentation, verify GitHub accessibility, and optionally add time-based validation for modelling robustness.

Scope and Objectives

The scope of the analysis is limited to program-level GES observations. The unit of analysis is a university-degree-year or program-year record. The project does not use individual graduate-

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level data and therefore does not make claims about individual student employability. All conclusions are at the program level.

The project aims to produce five measurable outputs:

1. a year-level trend summary of salary and employment outcomes from 2013 to 2024.
2. statistical evidence of whether university-level outcome differences are significant.
3. a regression-based explanation of factors associated with median salary and Career Value Index.
4. classification models for identifying high-employment, high-salary, and high-career-value program-year records.
5. clustering-based employability-salary profiles that can support student, university, and policy decision-making.

Research Question, Variables, Method, and Measurable Output Mapping

To address the capstone requirement for measurable outcomes, each research question is linked to specific variables, analytical methods, measurable outputs, and project deliverables. This ensures that the project is not only descriptive, but also produces measurable evidence for trend analysis, statistical comparison, regression, classification, and segmentation.

RQ1: Employment and Salary Trends

Research Question:

How have employment and salary outcomes evolved across Singapore university degree programs from 2013 to 2024?

Main variables from data dictionary:

year, employment_rate_overall, employment_rate_ft_perm, gross_monthly_median, salary_iqr, career_value_index

Method / solution approach:

Year-level trend analysis and descriptive exploratory data analysis.

Measurable output:

Median salary change from 2013 to 2024, yearly overall employment trend, yearly full-time permanent employment trend, and year-level Career Value Index pattern.

Capstone deliverable:

Year-level trend summary table, salary trend figure, employment trend figure, and RQ1 interpretation.

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RQ2: University-Level Outcome Differences

Research Question:

Do employment and salary outcomes differ significantly across universities and degree programs?

Main variables from data dictionary:

university, employment_rate_overall, employment_rate_ft_perm, gross_monthly_median, career_value_index

Method / solution approach:

Kruskal-Wallis statistical testing and post-hoc pairwise comparison.

Measurable output:

p-values, effect sizes, and evidence of statistically significant university-level differences.

Capstone deliverable:

Statistical test summary table, university comparison evidence, and RQ2 interpretation.

RQ3: Salary and Career-Value Drivers

Research Question:

Which factors contribute most to graduate salary outcomes and graduate career value across Singapore university programs?

Main variables from data dictionary:

year, employment_rate_overall, employment_rate_ft_perm, employment_premium, salary_iqr, gross_monthly_median, university, career_value_index

Method / solution approach:

Correlation analysis and linear regression.

Measurable output:

MAE, RMSE, R-squared, coefficient ranking, and correlation strength.

Capstone deliverable:

Salary regression model, Career Value Index diagnostic interpretation, and RQ3 driver analysis.

RQ4: High-Outcome Classification

Research Question:

Can machine learning models accurately predict high-employment, high-salary, and high-career-value degree programs?

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Main variables from data dictionary:

year, university, school, degree, and high-outcome labels derived from employment, salary, and Career Value Index thresholds.

Method / solution approach:

Logistic Regression, Decision Tree, Random Forest, majority-class baseline comparison, and leakage-controlled validation.

Measurable output:

Accuracy, precision, recall, F1-score, ROC AUC, and comparison against the baseline model.

Capstone deliverable:

High-employment, high-salary, and high-career-value classification results, with leakage-controlled model interpretation.

RQ5: Employability-Salary Segmentation

Research Question:

Can degree programs be segmented into meaningful employability-salary profiles to support student, university, and policy decision-making?

Main variables from data dictionary:

employment_rate_overall, employment_rate_ft_perm, gross_monthly_median, salary_iqr

Method / solution approach:

K-Means clustering and PCA visualization.

Measurable output:

Selected cluster count, silhouette score, PCA explained variance, and cluster profile medians.

Capstone deliverable:

Three employability-salary program profiles for decision support, including cluster interpretation for students, universities, and policymakers.

Literature Survey

Literature Review Approach

The literature review focused on graduate employability, machine learning model evaluation, regression, classification, clustering, and multivariate analysis. Sources were selected because they support the conceptual framing of employability, the statistical basis for the analysis, or the modelling techniques used in the notebook. The review also includes the official MOE dataset source because it is the primary data foundation for the project.

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Table 2 Literature relevance matrix

Author (Year)	Domain/Context	Method(s)	How it supports the project
MOE (2026)	Singapore graduate outcomes	Official open dataset	Primary data source for all RQs.
Tomlinson (2012)	Graduate employability	Conceptual review	Supports multidimensional view of employability.
Fugate et al. (2004)	Employability construct	Theoretical model	Supports interpreting employability beyond one metric.
Shmueli (2010)	Analytics purpose	Explain vs. predict framework	Guides distinction between explanatory regression and predictive classification.
James et al. (2023)	Statistical learning	Regression/classification methods	Supports modelling choices and validation metrics.
Breiman (2001)	Machine learning	Random forest	Supports tree-ensemble classification comparison.
Pedregosa et al. (2011)	Machine learning software	scikit-learn library	Supports implementation of models and pipelines.
MacQueen (1967)	Clustering	K-Means clustering	Supports segmentation method for RQ5.
Jolliffe & Cadima (2016)	Multivariate analysis	PCA review	Supports PCA visualization of clusters.
Green (1991)	Regression design	Sample-size guidance	Supports sample adequacy discussion.
Hair et al. (2019)	Multivariate statistics	Regression and multivariate methods	Supports correlation and multicollinearity interpretation.

The literature supports the project in three main ways. First, employability literature emphasizes that graduate outcomes are not limited to salary. Employment stability and career readiness must also be considered. This aligns with the project decision to include full-time permanent employment, salary spread, and a constructed Career Value Index.

Second, predictive analytics literature distinguishes between models used for explanation and models used for prediction. This distinction is important because the salary regression model is interpreted as an association model, while the classification models are evaluated using prediction metrics. The Career Value Index regression is treated as diagnostic because the index is constructed from its component variables.

Third, clustering and multivariate analysis literature supports the segmentation phase. K-Means clustering is appropriate for creating groups from standardized numeric outcome dimensions. PCA is used only for visualization, while cluster interpretation is based on the original employment and salary variables.

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Data Description

Data Source(s) and Access

The dataset used in this project is the Graduate Employment Survey - NTU, NUS, SIT, SMU, SUSS & SUTD dataset published by MOE through data.gov.sg. The dataset is made available under the Singapore Open Data Licence version 1.0. The project repository should include the raw file and processed outputs so that the evaluator can reproduce the analysis.

Dataset

URL:

https://data.gov.sg/datasets?resultId=d_3c55210de27fcccda2ed0c63fdd2b352&topics=education

Open Data Licence URL: <https://data.gov.sg/open-data-licence>

Dataset Overview

Table 3 Dataset overview

Item	Value
Number of records before cleaning	1,550
Number of variables before feature engineering	12
Number of variables after feature engineering	23
Complete-case records for key outcomes	1,445
Complete-case percentage	93.23%
Time period	2013 to 2024
Universities	NTU, NUS, SIT, SMU, SUSS, SUTD
Number of schools	78
Number of degree program names	419
Unit of analysis	University-degree-year / program-year record
Primary salary outcome	gross_monthly_median
Primary employment outcomes	employment_rate_overall, employment_rate_ft_perm
Composite outcome	career_value_index

Data Dictionary (Mandatory)

Table 4 Data dictionary

Variable Name	Definition	Datatype	Allowed Values/Range	Missing Values Handling Notes
year	Survey year	Integer	2013-2024	No missing values.
university	University name	Categorical	Six universities	No missing values.
school	School/faculty within university	Categorical	78 schools	No missing values.
degree	Degree program name	Categorical	419 program names	No missing values.

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Variable Name	Definition	Datatype	Allowed Values/Range	Missing Values Handling Notes
employment_rate_overall	Overall employment rate	Numeric	55.20-100.00	Missing retained as NaN; complete-case used for models.
employment_rate_ft_perm	Full-time permanent employment rate	Numeric	14.30-100.00	Missing retained as NaN.
gross_monthly_median	Median gross monthly salary	Numeric	1,800-7,000	Primary salary outcome.
gross_monthly_mean	Mean gross monthly salary	Numeric	2,100-7,244	Used for salary distribution checks.
gross_mthly_25_percentile	25th percentile gross monthly salary	Numeric	1,500-6,500	Used to calculate salary spread.
gross_mthly_75_percentile	75th percentile gross monthly salary	Numeric	2,500-9,967	Used to calculate salary spread.
salary_iqr	Salary interquartile range	Engineered numeric	0-5,817	Derived from 75th percentile minus 25th percentile.
salary_premium	Difference between gross mean and median salary	Engineered numeric	-491-2,139	Used to assess salary skew.
employment_premium	Gap between overall and full-time permanent employment	Engineered numeric	0-71.40	Used as employment stability indicator.
stability_score	Inverse normalized salary spread	Engineered numeric	0-1	Used as salary stability measure.
career_value_index	Composite index of employment, salary, and stability	Engineered numeric	0.31-0.97	Used as decision-support metric, not causal quality ranking.
analysis_complete_case	Flag for complete key outcomes	Boolean	0/1	Used to select model-ready records.

GitHub Data Availability Statement

The raw and processed datasets are available in the GitHub repository under the following paths:

- Raw data: data/raw/graduate_employment_survey.csv
- Processed/clean data: data/processed/ges_cleaned.csv
- Clustered data: data/processed/ges_clustered_program_profiles.csv
- Code/notebooks: notebooks/QM640_GES_Interim_Master_Analysis.ipynb

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- Tables: outputs/tables/
- Figures: outputs/figures/

```

QM640_GES_Capstone/
├── data/
│   ├── raw/
│   │   └── graduate_employment_survey.csv
│   └── processed/
│       ├── ges_cleaned.csv
│       └── ges_clustered_programme_profiles.csv
├── notebooks/
│   └── QM640_GES_Interim_Master_Analysis.ipynb
├── outputs/
│   ├── figures/
│   ├── tables/
│   └── report_package/
├── dashboard/
├── README.md
├── requirements.txt
└── LICENSE
  
```

Figure 1 Repository folder structure and dataset preview screenshot

The repository has been updated with the analysis notebook, processed datasets, output tables, report figures, report text files, and checkpoint metadata required to reproduce the interim analysis.

Minimum Sample Size Computation and Adequacy by Research Question

The dataset contains 1,550 program-year records before cleaning and 1,445 complete-case records for the main analysis. Because the project uses the full available population of program-level GES records rather than drawing a small sample, the available sample size is considered sufficient for descriptive, statistical, regression, classification, and clustering analysis.

For a conservative minimum sample-size check, the standard 95% confidence-level sample size formula was used:

$$n = Z^2 p (1 - p) / e^2$$

where $Z = 1.96$ for a 95% confidence level, $p = 0.50$ for maximum variability, and $e = 0.05$ for a 5% margin of error.

$$n = (1.96^2 \times 0.50 \times 0.50) / 0.05^2; n = 384.16 \sim 385$$

Therefore, the minimum recommended sample size is approximately 385 records. The complete-case dataset of 1,445 records is well above this minimum requirement.

For regression analysis, Green's rule of thumb was also considered:

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$$N \geq 50 + 8m$$

where m is the number of predictors. With approximately 8 candidate predictors, the minimum recommended sample size is:

$$N \geq 50 + 8(8) = 114$$

The complete-case dataset of 1,445 records is also well above this regression adequacy threshold.

Table 5 Sample Size Adequacy by Research Question

RQ	Analysis Type	Minimum Requirement Used	Available Records	Adequacy Decision
RQ1	Trend and descriptive EDA	385 records using 95% confidence sample-size check	1,445 complete cases	Adequate
RQ2	Group comparison / Kruskal-Wallis testing	385 records using 95% confidence sample-size check	1,445 complete cases	Adequate
RQ3	Regression and correlation	114 records using Green's rule with approximately 8 predictors	1,445 complete cases	Adequate
RQ4	Classification modelling	385 records and 362 high-outcome records per target	1,445 complete cases	Adequate
RQ5	Clustering and segmentation	385 records using 95% confidence sample-size check	1,445 complete cases	Adequate

Analysis

Data Cleaning

The data cleaning process preserved the original 1,550 program-year records. The main cleaning steps were standardizing column names, converting employment and salary columns from object type to numeric type, identifying non-numeric placeholders such as na and N.A., checking duplicate records, and documenting missing values. No duplicate rows were found.

The numeric conversion step identified 105 records with missing values across key employment and salary variables. These records represented 6.77% of the dataset. The missing values occurred as a complete block across the main outcome variables, so missing values were not imputed for the main analysis. Instead, descriptive analysis used available values and modelling used complete-case records. This resulted in 1,445 records for the main statistical and modelling phases.

Table 6 Data cleaning log

Issue	Variables Affected	Detection Method	Treatment Applied	Rationale
Non-numeric placeholders	Employment and salary fields	Pre-conversion value audit	Converted na, N.A., blanks, and similar placeholders to missing values	Prevents invalid numeric analysis.
Missing values	Key outcome variables	Missing-value summary by year and university	Retained as NaN and used complete-case analysis for models	Avoids artificial values in official outcome fields.
Duplicates	All columns	Duplicate-row check	No duplicate rows removed	Preserves original program-year records.
Outliers	Employment and salary variables	IQR and distribution checks	Retained for analysis	Outliers represent real program-level variation and are not data-entry errors.
Feature engineering	Salary and employment derived variables	Calculated from cleaned numeric fields	Created salary_iqr, employment_premium, stability_score, career_value_index	Supports multidimensional outcome analysis.

Exploratory Data Analysis Results

The exploratory data analysis (EDA) was conducted in four parts: coverage analysis, univariate analysis, bivariate trend and group-comparison analysis, and multivariate correlation analysis. The EDA confirmed that the dataset is suitable for trend analysis, group comparison, modelling, and segmentation.

The coverage analysis showed that the dataset contains 1,550 program-year records and 1,445 complete-case records. The dataset covers 2013 to 2024 and six universities. NTU, NUS, and SIT contribute the largest number of records, so comparisons must account for unequal group sizes. Missingness was moderate and concentrated in complete outcome blocks rather than scattered randomly across individual variables.

The year-level trend analysis showed that median gross monthly salary increased from 3,100 in 2013 to 4,400 in 2024. Overall employment remained generally high, but full-time permanent employment showed more variation, including a sharp decline in 2020 followed by recovery in 2021 and 2022. This finding supports the decision to analyze overall employment and stable employment separately.

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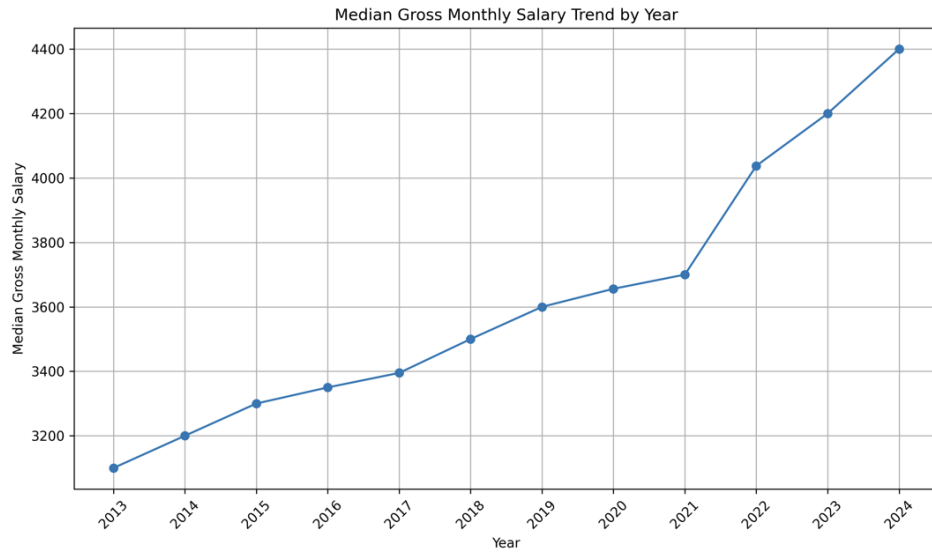


Figure 2 Median gross salary trend by year

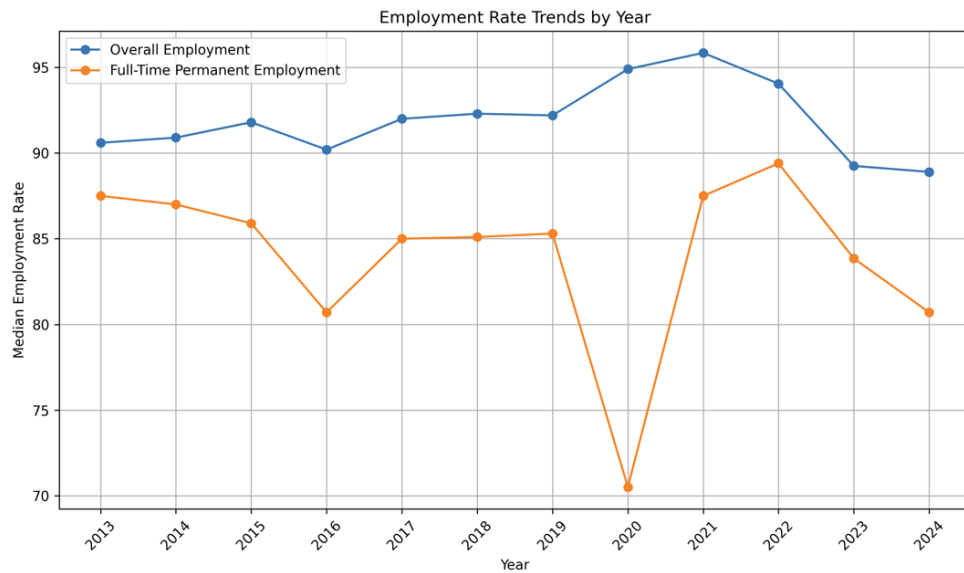


Figure 3 Overall and full-time permanent employment trends by year

University-level EDA showed visible differences in median salary, employment outcomes, and Career Value Index. SMU and SUTD had the highest median gross salary values at 4,000, while NUS had a median of 3,700 and NTU, SIT, and SUSS had medians of 3,500. SMU also showed the highest median Career Value Index at 0.76. These differences were later tested statistically.

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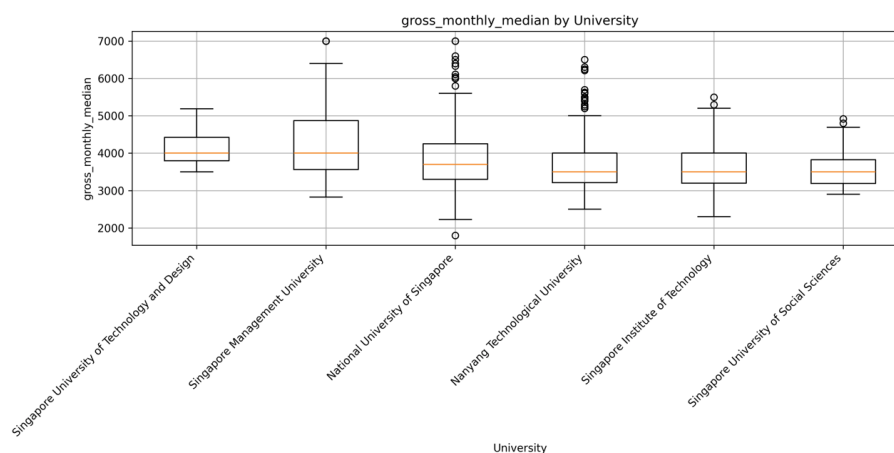


Figure 4 Median gross salary by university

The multivariate correlation analysis identified important relationships and redundancy. Career Value Index was strongly correlated with full-time permanent employment (0.91) and overall employment (0.86), and moderately correlated with median salary (0.52). Salary variables such as mean, median, 25th percentile, and 75th percentile were highly correlated. This guided feature selection for regression and classification.

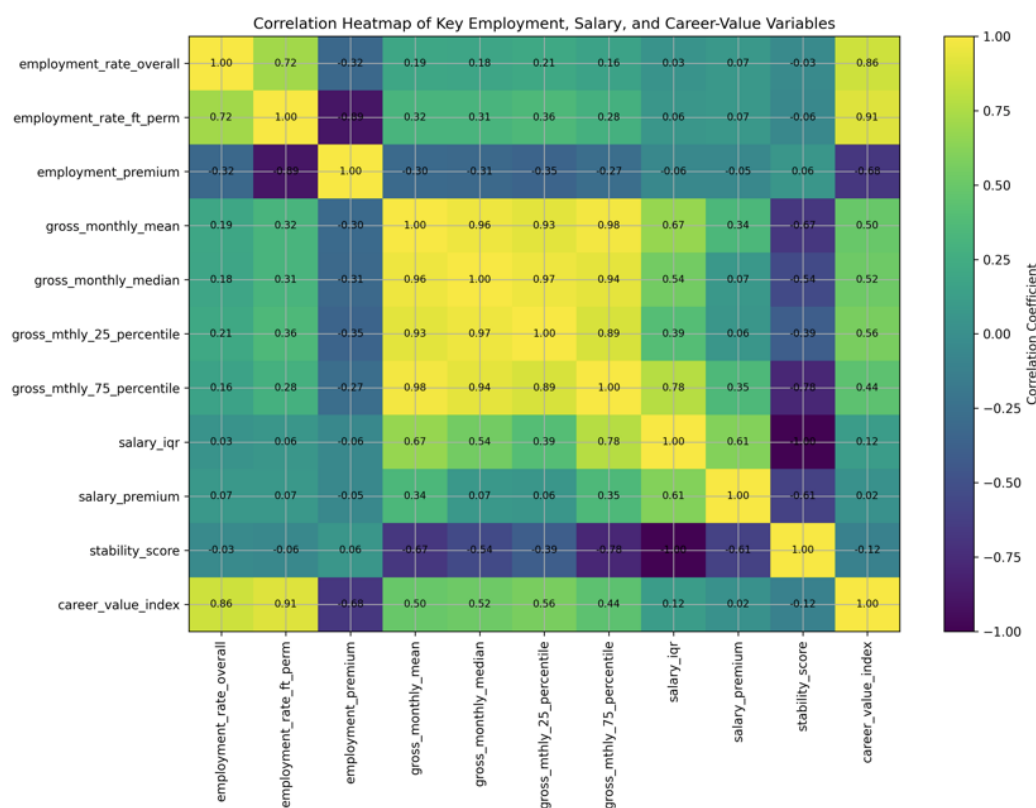


Figure 5 Correlation heatmap of employment, salary, and career-value variables

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Table 7 EDA insight summary

Area	Key Insight	Why it Matters	Next Step
Dataset coverage	1,550 records, 1,445 complete cases, 2013-2024 coverage	Supports trend and modelling readiness	Use complete cases for modelling.
Salary trend	Median salary increased from 3,100 to 4,400	Supports RQ1	Discuss salary growth over time.
Employment stability	Full-time permanent employment varied more than overall employment	Shows multidimensional outcomes	Analyze employment stability separately.
University differences	Visible differences in salary and Career Value Index	Supports RQ2	Apply Kruskal-Wallis testing.
Correlation structure	Career value strongly linked to employment stability	Supports RQ3	Use selected non-redundant variables in models.

Career Value Index Design and Validation

Career Value Index (CVI) is introduced as a constructed program-level decision-support index. The purpose of CVI is to summarize multiple graduate outcome dimensions into a single interpretable score so that program outcomes are not evaluated using salary alone. CVI combines overall employment, full-time permanent employment, median salary, and salary stability into one composite measure.

CVI formula

CVI = 0.30 Overall Employment Score+ 0.30 Full-Time Permanent Employment Score+ 0.25 Median Salary Score+ 0.15 Salary Stability Score

The CVI components are normalized before weighting so that employment percentages, salary values, and salary stability can be combined on a comparable scale. The theoretical CVI range is 0 to 1. A higher CVI indicates stronger combined program-level employment, stable employment, salary, and salary-stability outcomes. A lower CVI indicates weaker combined outcomes. In the interim dataset, observed CVI values range from 0.31 to 0.97.

Weighting rationale

The weights are design weights, not regression-derived coefficients. Overall employment and full-time permanent employment are each assigned 30% because employability is the central focus of the project. Median salary is assigned 25% so that salary remains an important outcome but does not dominate the index. Salary stability is assigned 15% to account for salary spread and consistency. Therefore, CVI gives 60% total weight to employment outcomes, 25% to salary level, and 15% to salary stability.

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CVI is not an official ranking, causal score, or individual employability predictor. It is a transparent program-level decision-support index constructed for this capstone analysis.

Table 8 Career Value Index Validation Evidence

Validation Feature	Expected Relationship	Pearson r	Pearson p-value	Spearman rho	Spearman p-value	Interpretation
Full-time permanent employment	Positive	0.91	p < .001	0.94	p < .001	Strong positive relationship; CVI strongly reflects stable employment.
Overall employment	Positive	0.86	p < .001	0.87	p < .001	Strong positive relationship; CVI reflects general employability.
Median gross monthly salary	Positive	0.52	p < .001	0.51	p < .001	Moderate positive relationship: CVI includes salary but is not salary-only.
Employment premium	Negative / caution	-0.68	p < .001	-0.74	p < .001	Moderate negative relationship: a larger gap between overall and full-time permanent employment is associated with lower CVI.
Salary stability score	Positive	-0.12	p < .001	-0.09	p < .001	Weak relationship: direction should be reviewed further through sensitivity analysis.
Salary IQR	Negative	0.12	p < .001	0.09	p < .001	Weak relationship: wider salary spread may appear in some higher-CVI programs, so the effect is weaker than employment and salary level.

The Career Value Index validation evidence shows that CVI is statistically aligned with a meaningful set of graduate outcome features. CVI has strong positive and statistically significant relationships with full-time permanent employment and overall employment, and a moderate positive and statistically significant relationship with median gross monthly salary. Employment premium also shows a moderate negative and statistically significant relationship with CVI, indicating that a larger gap between overall employment and full-time permanent employment is associated with lower career value. These results support the use of CVI as a representative program-level summary index because multiple relevant graduate outcome features are significantly related to CVI. However, the salary stability score and salary IQR show weak relationships, so the stability component should be reviewed further through sensitivity analysis. Because CVI is constructed from normalized employment, salary, and stability components, these correlation results are interpreted as consistency validation rather than independent causal proof. Therefore, CVI is used as a transparent program-level decision-support index, not as an official ranking, causal score, or individual employability predictor.

Planned Career Value Index Sensitivity Analysis

To strengthen the Career Value Index validation in the final report, a sensitivity analysis will be conducted using alternative weighting schemes. The purpose is to test whether the CVI interpretation remains stable when the weights assigned to employment, salary, and salary stability are adjusted. This will help assess whether the proposed CVI is robust and not overly dependent on one weighting structure.

Table 9 Planned CVI Sensitivity Weighting Schemes

CVI Version	Overall Employment Score	Full-Time Permanent Employment Score	Median Salary Score	Salary Stability Score	Purpose
Current CVI	30%	30%	25%	15%	Main proposed index
Equal-weight CVI	25%	25%	25%	25%	Tests balanced weighting
Employment-heavy CVI	35%	35%	20%	10%	Tests stronger employability emphasis
Salary-heavy CVI	25%	25%	35%	15%	Tests stronger salary emphasis
Stability-heavy CVI	25%	25%	20%	30%	Tests stronger salary-consistency emphasis

In the final report, the current CVI will be compared with these alternative CVI versions using Spearman rank correlation, top-quartile overlap, and cluster-profile stability. If the alternative CVI versions remain highly correlated with the current CVI and produce similar cluster interpretations, this will provide additional support for CVI as a robust program-level decision-support index.

Modelling

Choice of Models with Justification

The modelling phase included statistical testing, regression, classification, leakage-controlled validation, and clustering. Kruskal-Wallis testing was selected as the primary group-comparison method because EDA showed skewness, outliers, and unequal variances. Linear regression was used for interpretable association modelling. Logistic Regression, Decision Tree, and Random Forest classifiers were used for high-outcome classification. K-Means clustering was used for segmentation because the goal was to create interpretable profiles from standardized numeric outcome variables.

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The modelling workflow was organized into five phases: Phase 1 data preparation and feature engineering, Phase 2 EDA, Phase 3 statistical testing and predictive modelling, Phase 4 clustering and segmentation, and Phase 5 interim findings consolidation and reporting preparation.

Table 10 Feature set and usage

Feature	Original/Engineered	Type	Reason for Inclusion	Used in Model(s)
year	Original	Numeric	Captures time trend	Regression, classification
university	Original	Categorical	Captures institutional context	Regression, classification
school	Original	Categorical	Program context	Leakage-controlled classification
degree	Original	Categorical	Program identifier/context	Leakage-controlled classification
employment_rate_overall	Original	Numeric	Primary employment outcome	EDA, regression, clustering
employment_rate_ft_perm	Original	Numeric	Employment stability outcome	EDA, regression, clustering
gross_monthly_median	Original	Numeric	Primary salary outcome	EDA, regression, clustering
salary_iqr	Engineered	Numeric	Salary spread indicator	Regression, clustering
employment_premium	Engineered	Numeric	Gap between overall and full-time permanent employment	Regression, preliminary classification
career_value_index	Engineered	Numeric	Composite employment-salary-stability index	Diagnostic model and cluster interpretation

Evaluation Metrics (Formulae and Calculations)

Table 11 Evaluation metrics and formulae

Metric	Formula	Interpretation	Used For
Accuracy	$(TP + TN) / (TP + TN + FP + FN)$	Overall classification correctness	Classification
Precision	$TP / (TP + FP)$	Share of predicted high-outcome cases that are	Classification

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Metric	Formula	Interpretation	Used For
		correct	
Recall	$TP / (TP + FN)$	Share of actual high-outcome cases correctly found	Classification
F1-score	$2 \times (Precision \times Recall) / (Precision + Recall)$	Balance between precision and recall	Classification
ROC AUC	Area under ROC curve	Ranking ability across thresholds	Classification
MAE	$\text{mean}(y - \hat{y})$	Average absolute prediction error	Regression
RMSE	$\sqrt{\text{mean}((y - \hat{y})^2)}$	Prediction error with stronger penalty for large errors	Regression
R-squared	$1 - SS_{\text{res}} / SS_{\text{tot}}$	Proportion of target variation explained by the model	Regression
Silhouette score	$(b - a) / \max(a, b)$	Cluster separation and cohesion	Clustering

Preliminary Results

RQ1: Employment and Salary Trends

RQ1: How have employment and salary outcomes evolved across Singapore university degree programs from 2013 to 2024?

The year-level trend analysis shows that program-level salary outcomes increased steadily across the study period. Median gross monthly salary increased from **3,100 in 2013** to **4,400 in 2024**, indicating a clear upward trend in graduate salary outcomes. This pattern is shown in **Figure 2** and is supported by the year-level trend summary generated from `outputs/tables/bivariate_year_trend_summary.csv`.

Table 12 Bivariate year and trend summary

year	record_count	median_ overall_employment	median_ ft_perm_employment	median_ gross_salary	median_ salary_iqr	median_ career_value_index
2013	85	90.6	87.5	3100	600	0.687817832
2014	109	90.9	87	3200	600	0.696882823
2015	123	91.8	85.9	3300	625	0.699109257
2016	121	90.2	80.7	3350	663	0.674230295

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year	record_count	median_ overall_employment	median_ ft_perm_employment	median_ gross_salary	median_ salary_iqr	median_ career_value_index
2017	132	92	85	3395	700	0.691748734
2018	137	92.3	85.1	3500	710	0.707400188
2019	140	92.2	85.3	3600	700	0.714660783
2020	139	94.9	70.5	3656	772	0.682846283
2021	135	95.85	87.5	3700	745	0.752339505
2022	141	94.05	89.4	4037.5	950	0.760847399
2023	139	89.25	83.85	4200	835.5	0.712266103
2024	149	88.9	80.7	4400	900	0.705045701

Employment outcomes remained generally high across most years, but overall employment and full-time permanent employment did not move identically. Median overall employment reached 95.85% in 2021, while full-time permanent employment showed greater variation. The sharpest decline in full-time permanent employment occurred in 2020, when the median dropped to 70.50%, before recovering in 2021 and 2022. This trend is shown in Figure 3.

The RQ1 findings show that salary outcomes improved over time, while employment stability fluctuated more than overall employment. This confirms that graduate outcomes should not be interpreted using salary alone. Overall employment, full-time permanent employment, salary level, and salary stability should be considered together when evaluating program-level outcomes.

RQ2 : Statistical Testing Results

RQ2: Do employment and salary outcomes differ significantly across universities and degree programs?

Kruskal-Wallis tests found statistically significant university-level differences for all four tested outcomes: overall employment rate, full-time permanent employment rate, median gross monthly salary, and Career Value Index. Because the p-values were extremely small and displayed as 0.00 due to rounding, they should be reported as $p < .001$ rather than $p = 0$. The effect sizes were modest, with epsilon-squared values from approximately 0.05 to 0.07.

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Table 13 Statistical test summary

Outcome	Kruskal-Wallis Result	Effect Size	Interpretation
Overall employment rate	Statistically significant, $p < .001$	0.05	University-level differences exist, but effect size is modest.
Full-time permanent employment rate	Statistically significant, $p < .001$	0.06	Stable employment differs across universities.
Median gross monthly salary	Statistically significant, $p < .001$	0.06	Salary outcomes differ across universities.
Career Value Index	Statistically significant, $p < .001$	0.07	Composite career-value profiles differ across universities.

RQ3 : Regression Results

RQ3: Which factors contribute most to graduate salary outcomes and graduate career value across Singapore university programs?

The salary regression model used 1,445 complete-case records with a 75:25 train-test split. The model achieved MAE = 333.86, RMSE = 476.90, and R-squared = 0.58. Important associated factors included year, salary spread, employment premium, full-time permanent employment, and university context. These results are associational and should not be interpreted causally.

The Career Value Index regression produced R-squared = 1.00 because the model included variables used to construct the index. This result is not treated as independent predictive evidence. Instead, it is used as a diagnostic validation showing that the index behaves consistently with its employment, salary, and stability components.

Table 14 Regression performance summary

Model	Target	Records Used	MAE	RMSE	R-squared	Key Interpretation
Salary regression	gross_monthly _median	1,445	333.86	476.90	0.58	Reasonable interim-stage association model for salary outcomes.
Career Value Index diagnostic regression	career_value_index	1,445	0.00	0.00	1.00	Diagnostic only; not independent prediction because of component overlap.

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RQ4 : Classification Results

RQ4: Can machine learning models accurately predict high-employment, high-salary, and high-career-value degree programs?

Preliminary classification models were developed for high employment, high salary, and high Career Value Index using top-quartile thresholds. The preliminary models performed very strongly, but the results were likely optimistic because some engineered variables indirectly contained target information. To address this risk, leakage-controlled classification was run using only year, university, school, and degree as predictors.

The leakage-controlled models produced more realistic results. High-salary classification was the strongest conservative task. Logistic Regression achieved $F1 = 0.79$ and $ROC\ AUC = 0.96$ under the random stratified context-only split. Under the stricter degree-grouped context-only split, the high-salary model achieved $F1 = 0.71$ and $ROC\ AUC = 0.91$. These results support RQ4 while demonstrating the importance of feature leakage control.

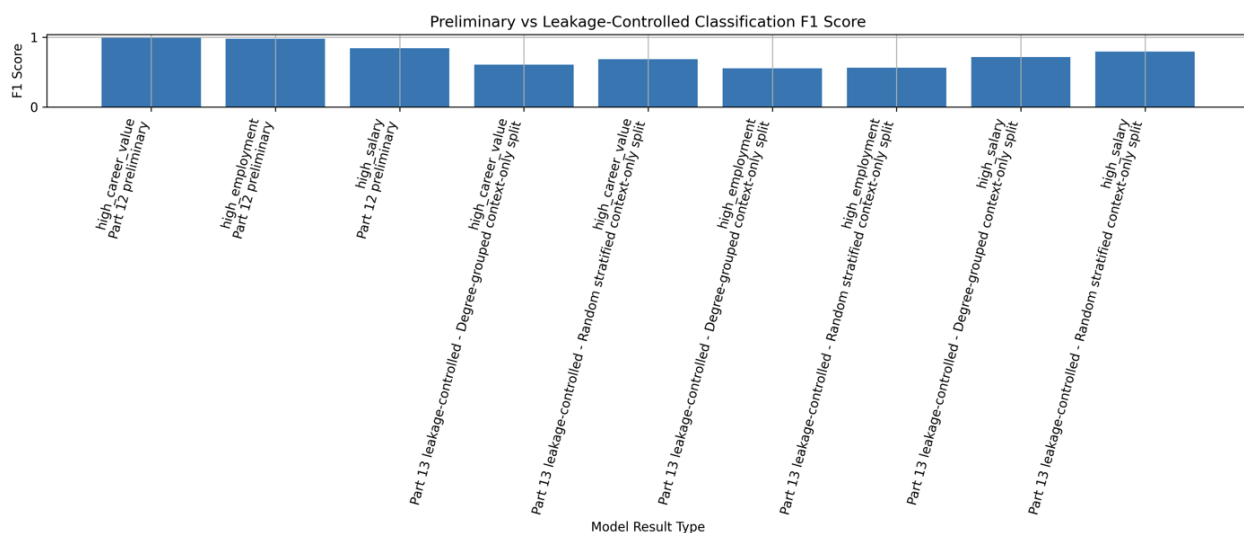


Figure 6 Preliminary versus leakage-controlled classification F1 scores

Table 15 Preliminary and leakage-controlled classification performance

Target	Best Preliminary F1	Best Leakage-Controlled F1	Best Conservative Validation	Key Takeaway
High employment	0.98	0.55-0.56	Logistic Regression, context-only	Moderate performance after leakage control.
High salary	0.84	0.71-0.79	Logistic Regression, context-only	Strongest conservative classification result.
High Career Value	0.99	0.61-0.69	Logistic Regression, context-only	Moderate performance after leakage control.

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RQ 5: Clustering Results

RQ5: Can degree programs be segmented into meaningful employability-salary profiles to support student, university, and policy decision-making?

K-Means clustering was used to segment program-year records into employability-salary profiles. The clustering inputs were overall employment rate, full-time permanent employment rate, median gross monthly salary, and salary IQR. Career Value Index was not used as a clustering input; it was used only after clustering to interpret the resulting profiles.

K values from 2 to 8 were evaluated. The highest silhouette score was observed at $k = 3$, with a silhouette score of 0.32. PCA showed that the first two principal components explained 82% of the variation in the clustering features. The three-cluster solution was selected because it provided interpretable and reportable outcome profiles.

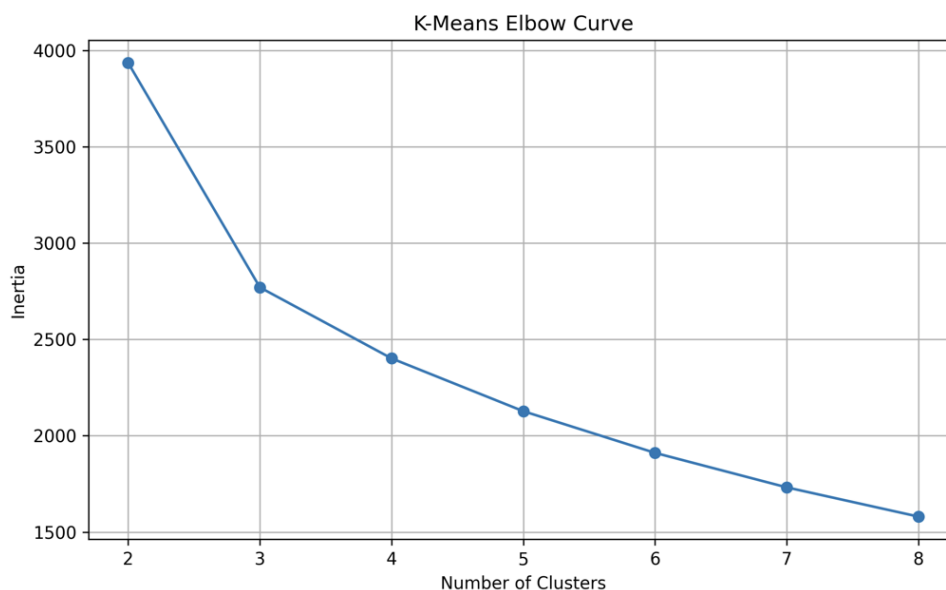


Figure 7 K-Means elbow curve

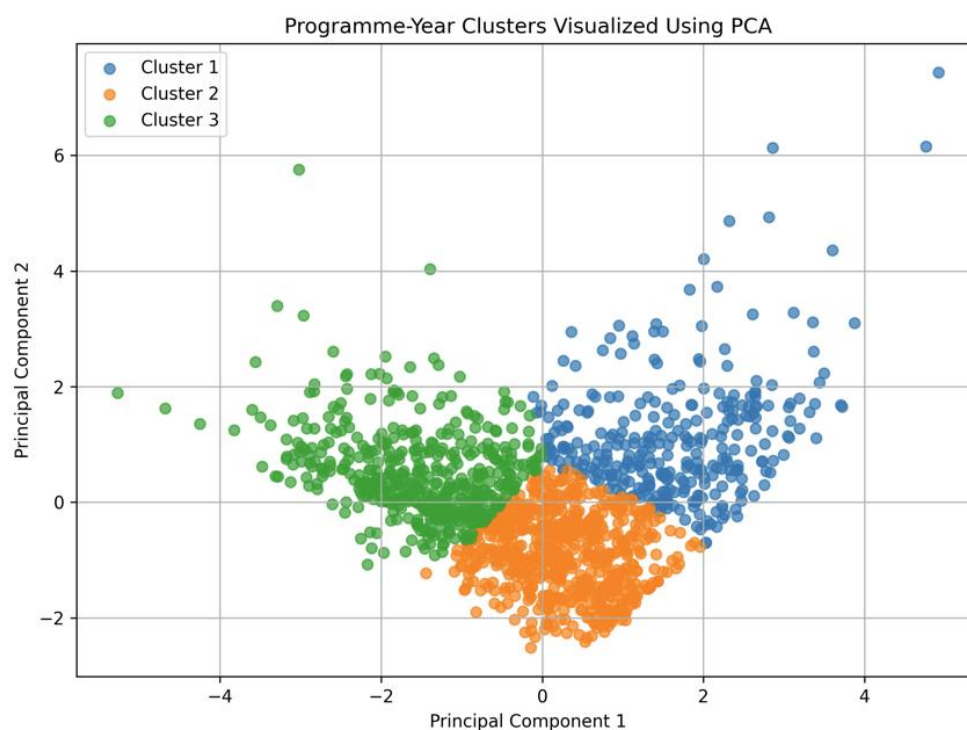


Figure 8 Program-year clusters visualized using PCA

The cluster profiles provide practical decision-support value because they translate multiple quantitative indicators into interpretable program outcome groups. Students can use the profiles to compare programs beyond salary alone. Universities can use the profiles to identify programs with strong employment but moderate salary, or programs with weaker full-time permanent employment. Policymakers can use the segmentation framework as a repeatable monitoring tool when new GES data is released.

Table 16 Cluster profile summary

Cluster	Record Count	Record %	Median Overall Employment	Median FT Permanent Employment	Median Salary	Median Salary IQR	Median CVI	Interpretation
1	324	22.42%	94.90%	91.10%	4,781.50	1,300	0.79	High employment and high salary profile
2	651	45.05%	94.90%	89.20%	3,500	600	0.74	High employment stability with moderate salary profile
3	470	32.53%	84.75%	68.25%	3,362.50	700	0.60	Moderate or lower relative outcome profile

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Preliminary Findings by Research Question

Table 17 summarizes the interim response to each research question and links the response to the technical evidence generated from the analysis. This structure helps connect the project findings to measurable outputs, including trend results, statistical tests, regression metrics, classification results, and clustering evidence. All findings are interpreted at the program-year level and should not be treated as individual student predictions or causal claims.

Table 17 Preliminary Research Question Responses and Technical Evidence

Research Question	Response	Technical Evidence	Final Report Action
RQ1: Are graduate salary and employment outcomes changing over time?	Yes. Salary increased over time, but stable employment varied more than overall employment.	Median gross salary increased from 3,100 in 2013 to 4,400 in 2024 . Full-time permanent employment dropped to 70.50% in 2020 before recovering.	Strengthen interpretation of salary growth and the 2020 employment-stability decline.
RQ2: Are outcomes different across universities?	Yes. Universities show statistically significant differences, but the differences are modest.	Kruskal-Wallis tests were significant for overall employment, full-time permanent employment, median salary, and Career Value Index. Effect sizes were approximately 0.05 to 0.07 .	Report only meaningful pairwise university differences in the final report.
RQ3: What factors are linked to salary and career value?	Salary is associated with year, salary spread, employment stability, employment gap, and university context.	Salary regression achieved MAE = 333.86 , RMSE = 476.90 , and R² = 0.58 . Important associated factors included year, salary spread, employment premium, full-time permanent employment, and university context.	Add clearer coefficient interpretation and strengthen CVI validation.
RQ4: Can high-outcome programs be predicted?	Yes, but high-salary programs are predicted more reliably than high-employment or high-CVI programs.	Leakage-controlled high-salary classification achieved F1 = 0.71 and ROC AUC = 0.91 under the stricter degree-grouped context-only split.	Use leakage-controlled results as the main predictive claim.
RQ5: Can programs be grouped into useful outcome profiles?	Yes. Three interpretable employability-salary profiles were identified.	K-Means selected k = 3 , with silhouette score 0.32 . Cluster median CVI values were 0.79 , 0.74 , and 0.60 .	Expand decision-support interpretation for students, universities, and policymakers.

All preliminary findings are interpreted as program-level associations and descriptive patterns; the analysis does not claim that university, degree, or year causes an individual graduate outcome.

Interim Limitations and Risks

Table 18 Limitations and mitigation plan

Area	Limitation/Risk	Impact	Mitigation
Data granularity	Dataset is program-year level, not individual-level.	Individual predictions cannot be claimed.	State all findings as program-level.
Causality	Analysis is observational.	Regression and classification are associational.	Avoid causal language.
Missing values	105 records missing key outcomes.	Models use 1,445 complete cases.	Report complete-case handling clearly.
Class imbalance	High-outcome class is about 25%.	Accuracy alone can mislead.	Report precision, recall, F1, ROC AUC, and baseline.
Feature leakage	Preliminary classification used related engineered variables.	Part 12 results are optimistic.	Use Part 13 leakage-controlled results as main evidence.
Clustering interpretation	Clusters are descriptive profiles.	Profiles are not causal rankings.	Use clusters for decision support only.

Next Steps for Final Report

- Strengthen the **RQ1 trend analysis** by adding clearer interpretation of salary growth, employment stability variation, and the 2020 full-time permanent employment decline.
- Expand the **RQ2 university-level comparison** by discussing only the most meaningful pairwise differences from the Kruskal-Wallis and post-hoc test results.
- Refine the **RQ3 regression interpretation** by explaining how year, salary spread, employment premium, full-time permanent employment, and university context are associated with median gross salary.
- Use the **leakage-controlled classification results** as the main evidence for **RQ4**, especially the high-salary classification model, because it provides the most defensible predictive result.
- Clearly separate **preliminary classification results** from **leakage-controlled results** so that optimistic model performance is not overclaimed.
- Strengthen the **RQ5 clustering discussion** by explaining how the three program profiles support student, university, and policy decision-making.
- Add a stronger final discussion showing that graduate outcomes are **multidimensional** and should be evaluated using salary, overall employment, full-time permanent employment, salary spread, and Career Value Index together.
- Review all findings to ensure the final report avoids causal claims and presents the results as program-level, data-driven evidence.
- Prepare the final recommendations based on the five research questions, with separate implications for students, universities, and policymakers.

Recommendations and Application

The analysis supports a multidimensional approach to graduate outcome interpretation. Students should not compare programs using salary alone. Instead, they should consider salary level, overall employment, full-time permanent employment, salary spread, and career-value profile together. A program with moderate salary and high stable employment may be preferable for some students compared with a program with high salary but wider spread.

Universities can use the findings to monitor programs that fall into the moderate or lower relative outcome cluster, especially when full-time permanent employment is low. The cluster profiles can also help identify programs that have strong employment but moderate salary outcomes, which may benefit from career-services support, employer engagement, or clearer communication about graduate pathways.

Policymakers can use this style of analysis as a monitoring framework. Because the data is already publicly available, a repeatable analytics workflow can be updated when new GES years are released. The leakage-controlled classification and clustering results are useful for decision support but should not be used as direct program rankings.

Overall Interim Conclusion

The interim analysis shows that graduate outcomes in Singapore should be interpreted as multidimensional program-level outcomes rather than as salary-only indicators. Salary outcomes increased from 2013 to 2024, but employment stability varied across years. University-level differences were statistically significant but modest in effect size, suggesting that university context matters but does not fully explain graduate outcomes. Regression results showed that salary outcomes are associated with year, salary spread, employment premium, full-time permanent employment, and university context. Classification results demonstrated that high-salary outcomes can be identified more reliably than high-employment or high-career-value outcomes under leakage-controlled validation. Finally, clustering identified three interpretable employability-salary profiles, supporting the use of segmentation for decision support.

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Appendix A: Report Output and Figure/Table Path Guide

The appendix lists the main output files generated by the notebook. These files should be used to paste figures and tables into the final formatted report. The appendix is referenced in the Analysis and Preliminary Results sections.

Table A1 Key figure paths

Figure	File Path	Report Section
Figure 1	GitHub repository screenshot	GitHub Data Availability Statement
Figure 2	outputs/figures/figure_07_01_salary_trend_by_year.png	EDA Results / RQ1
Figure 3	outputs/figures/figure_07_02_employment_trend_by_year.png	EDA Results / RQ1
Figure 4	outputs/figures/figure_10_boxplot_gross_monthly_median_by_university.png	Statistical Testing / RQ2
Figure 5	outputs/figures/figure_08_01_correlation_heatmap.png	Multivariate Analysis / RQ3
Figure 6	outputs/figures/figure_13_02_preliminary_vs_leakage_controlled_f1.png	Classification / RQ4
Figure 7	outputs/figures/figure_14_01_kmeans_elbow_curve.png	Clustering / RQ5
Figure 8	outputs/figures/figure_14_03_cluster_pca_scatter.png	Clustering / RQ5
Figure 9	outputs/figures/figure_14_04_cluster_profile_employment_career_value.png	Cluster profile interpretation
Figure B1	outputs/figures/appendix/figure_B1_combined_univariate_eda_summary.png	Appendix B / Univariate EDA
Figure B2	outputs/figures/appendix/figure_B2_combined_bivariate_eda_summary.png	Appendix B / Bivariate EDA
Figure B3	outputs/figures/appendix/figure_B3_multivariate_cvi_validation_summary.png	Appendix B / Multivariate and CVI validation

Table A2 Key table paths

Table Output	File Path	Use
Dataset overview	outputs/tables/dataset_overview.csv	Data Description
Data dictionary	outputs/tables/data_dictionary.csv	Data Dictionary
Data cleaning log	outputs/tables/data_cleaning_log.csv	Data Cleaning
Year trend summary	outputs/tables/bivariate_year_trend_summary.csv	RQ1
University outcome summary	outputs/tables/bivariate_university_outcome_summary.csv	RQ2

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Table Output	File Path	Use
Statistical test results	outputs/tables/statistical_test_results.csv	RQ2
Regression performance	outputs/tables/regression_performance_summary.csv	RQ3
Classification comparison	outputs/tables/classification_preliminary_vs_leakage_controlled.csv	RQ4
Cluster profile	outputs/tables/cluster_profile_simple.csv	RQ5
CVI correlation validation	outputs/tables/cvi_correlation_validation.csv	CVI validation / RQ3
Final RQ summary	outputs/tables/final_research_question_results.csv	Findings summary
Appendix B figure index	outputs/tables/appendix_b_figure_index.csv	Appendix figure tracking

Appendix B: Detailed Exploratory Data Analysis Outputs

This figure summarizes the distribution of key employment, salary, salary-spread, and Career Value Index variables used in the analysis.

Figure B1. Combined Univariate EDA Summary

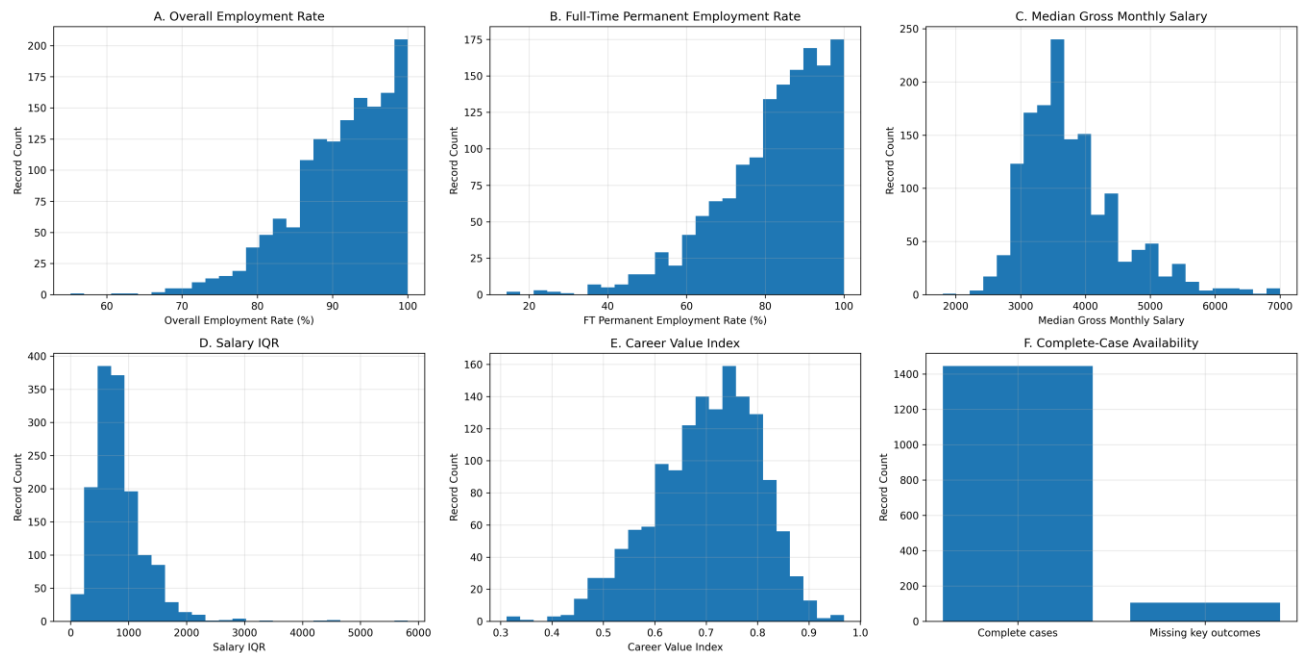
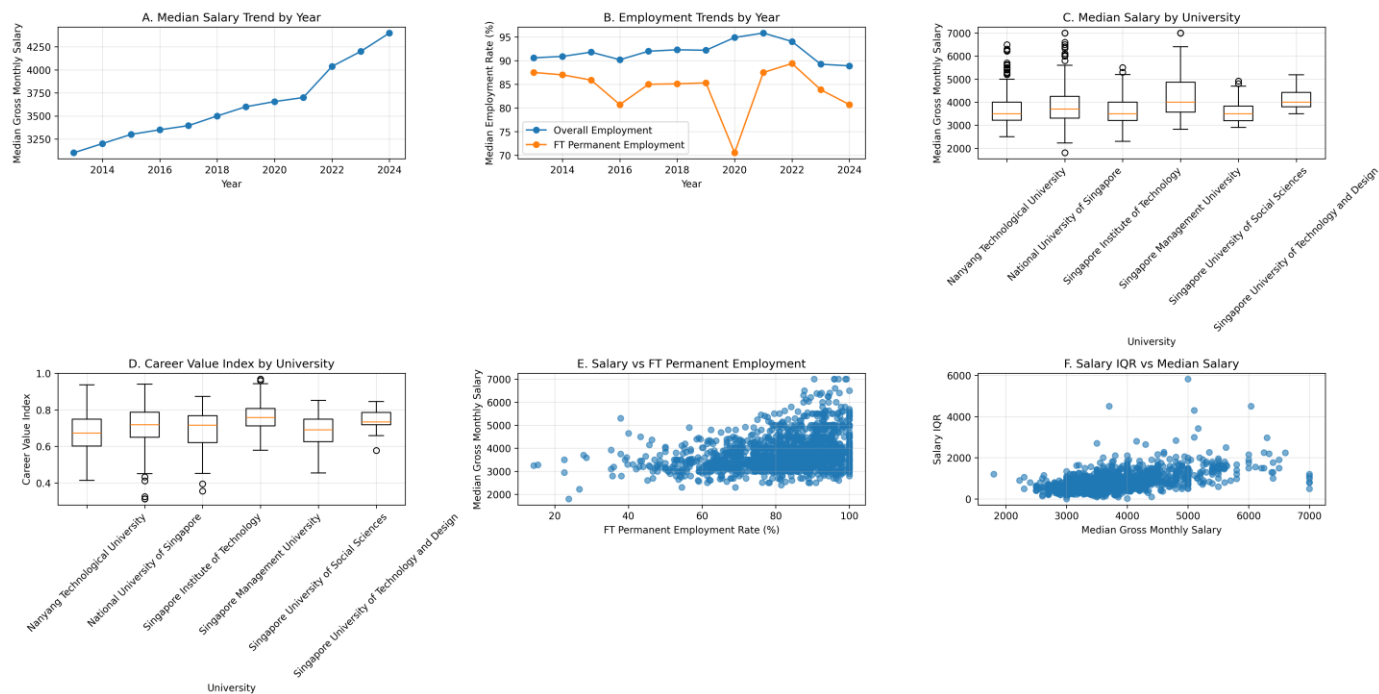


Figure B1: Combined Univariate EDA Summary

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This figure summarizes the showing year trends, university differences, and relationships between salary, employment, and salary spread.

Figure B2. Combined Bivariate EDA Summary

**Figure B2: Combined Bivariate EDA Summary**

This figure supports RQ1, RQ2, and RQ3 by showing trends, group-level differences, and relationships between major outcome variables.

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This figure summarizes the Multivariate and Career Value Index validation summary showing correlation structure, CVI feature relationships, and cluster-level CVI interpretation.

Figure B3. Multivariate and CVI Validation Summary

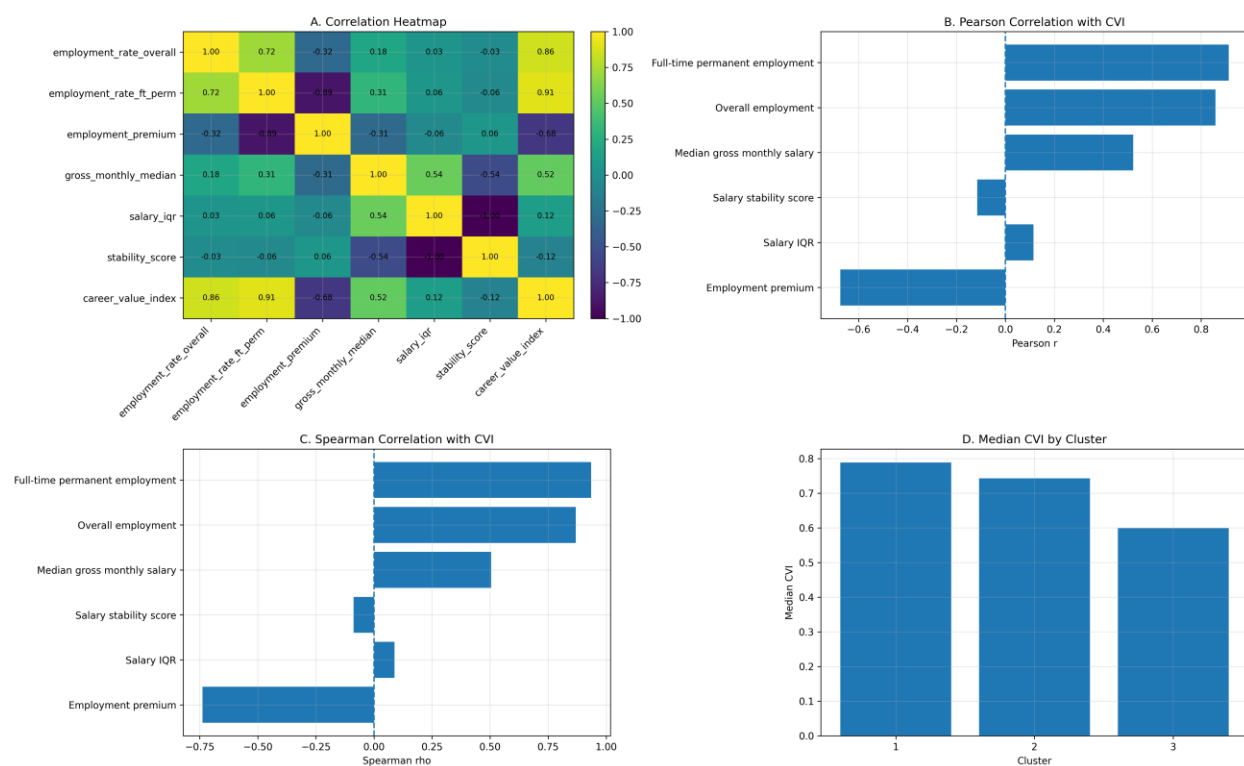


Figure B3: Multivariate and CVI Validation Summary

This figure supports the statistical validation of CVI as a representative program-level summary index.